API Request Log Analysis Report

Overview

This document summarizes API activity captured during testing of the IAM platform. The objective is to establish a baseline for endpoint performance, identify potential bottlenecks, and provide historical data for future optimization efforts.

Test Date: 17 June 2026
Environment: Local Development
Base URL: http://localhost:8080

Endpoint Performance Summary

Endpoint	Method	Status	Response Time (ms)	Success
/api/v1/auth/signup	POST	201	7520	Yes
/api/v1/auth/login	POST	200	3486	Yes
/api/v1/user/me	GET	200	220	Yes
/api/v1/user/update	PUT	500	70	No
/api/v1/user/me/update	PUT	200	55	Yes
/api/v1/user/me/logs	GET	200	150	Yes
/api/v1/user/my-logs/download	GET	200	924	Yes
/api/v1/user/me/profile/download	GET	200	53	Yes

Detailed Observations

1. User Registration

Endpoint: POST /api/v1/auth/signup

- Response Time: 7520 ms
- Status: 201 Created

Observation:

The signup operation is currently the slowest endpoint in the system. A response time of over 7 seconds indicates expensive processing during account creation.

Potential causes:

- Password hashing cost factor
- Database insert latency
- User validation checks
- Role assignment logic
- Audit log creation
- Transaction overhead

Recommendation:

Profile the signup flow and measure the execution time of each internal service call separately.

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2. User Login

Endpoint: POST /api/v1/auth/login

- Response Time: 3486 ms
- Status: 200 OK

Observation:

Login latency is significantly higher than expected for a local environment.

Potential causes:

- BCrypt password verification
- JWT generation
- Session creation
- Audit logging
- Database user lookup

Recommendation:

Add timing logs around:

- User retrieval
- Password verification
- JWT creation
- Session persistence

to identify the dominant cost.

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3. User Profile Retrieval

Endpoint: GET /api/v1/user/me

- Response Time: 220 ms
- Status: 200 OK

Observation:

Acceptable performance for authenticated profile retrieval.

Recommendation:

No immediate optimization required.

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4. Failed Update Request

Endpoint: PUT /api/v1/user/update

- Response Time: 70 ms
- Status: 500 Internal Server Error

Observation:

The endpoint failed quickly, suggesting the error occurred before significant business processing.

Recommendation:

Review application logs and exception traces for root-cause analysis.

Potential causes include:

- Validation failures
- Null pointer exceptions
- Database constraint violations
- Mapping errors

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5. User Self Update

Endpoint: PUT /api/v1/user/me/update

- Response Time: 55 ms
- Status: 200 OK

Observation:

Excellent response time.

Recommendation:

Current performance is satisfactory.

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6. User Log Retrieval

Endpoint: GET /api/v1/user/me/logs

- Response Time: 150 ms
- Status: 200 OK

Observation:

Good performance.

Recommendation:

Monitor growth of audit log records and introduce pagination if needed.

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7. Audit Log Download

Endpoint: GET /api/v1/user/my-logs/download

- Response Time: 924 ms
- Status: 200 OK

Observation:

Download generation takes nearly one second.

Potential causes:

- File generation
- Data serialization
- Database retrieval
- Stream creation

Recommendation:

Benchmark:

- Database fetch time
- File generation time

- Response streaming time

individually.

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8. Profile Export

Endpoint: GET /api/v1/user/me/profile/download

- Response Time: 53 ms
- Status: 200 OK

Observation:

Fastest endpoint in the test set.

Recommendation:

No optimization required.

Performance Ranking

Rank	Endpoint	Response Time (ms)
1	/api/v1/auth/signup	7520
2	/api/v1/auth/login	3486
3	/api/v1/user/my-logs/download	924
4	/api/v1/user/me	220
5	/api/v1/user/me/logs	150
6	/api/v1/user/update (failed)	70
7	/api/v1/user/me/update	55
8	/api/v1/user/me/profile/download	53

Key Findings

1. Authentication endpoints are the primary performance bottleneck.
2. Signup latency exceeds acceptable thresholds for production workloads.
3. Login latency requires further investigation.
4. Download endpoints perform reasonably but should be monitored as data volume increases.
5. User management endpoints demonstrate strong response times.

6. One endpoint produced an internal server error and requires debugging.

Future Optimization Plan

Phase 1 – Measurement

Add execution-time logging around:

- Database operations
- Password hashing
- JWT generation
- Session creation
- File generation

Phase 2 – Profiling

Measure:

- Service layer execution time
- Repository execution time
- Security filter execution time

Phase 3 – Optimization

Potential improvements:

- Database indexing
- Query optimization
- Reduced BCrypt cost (if appropriate)
- Async audit logging
- Response caching where applicable

Baseline Metrics (June 2026)

Average Response Time: 1560 ms

Fastest Endpoint: GET /api/v1/user/me/profile/download (53 ms)

Slowest Endpoint: POST /api/v1/auth/signup (7520 ms)

System Success Rate: 87.5% (7 successful requests out of 8)